

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Stress-related illness: ulcers, high blood pressure, heart disease (1) • Behavioural changes: irritability, aggression, lack of concentration leading to accidents (1) • Communication problems / educational underperformance (eg schools under flight paths) (1) • Sleep disturbance (1)	3
01.2	<i>1 mark for each point.</i> • Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) (1) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes (1) • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry (1)	3
01.3	<i>1 mark for each point.</i> • Birds that sing to establish territories / attract mates cannot be heard against background noise (1) • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea (1)	2
01.4	<i>1 mark for each point.</i> • Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source (1) • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
02.1	<i>Correct answer only.</i> • B - Steam washing (hot water / steam jets kill organisms in the sediment) (1)	1
02.2	<i>1 mark for each point.</i> • Chemicals cause the oil molecules to join together / solidify (1) • Producing more solid material that can be collected / lifted more easily than liquid oil (1)	2
02.3	<i>1 mark for each point.</i> • Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) (1) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
03.1	<i>1 mark for each point.</i> • CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again (1) • During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls (1) • In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises (1) • The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large (1)	4
03.2	<i>1 mark for each point.</i> • Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) (1) • So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean (1)	2
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) (1) • Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) (1) • Atmospheric pressure allows liquid water to exist (1) • Distributes heat from the tropics to higher latitudes by winds (1) • Transports water vapour to areas that would otherwise be dry (1)	2
03.4	<i>1 mark for each point.</i> • Troposphere (1) • Stratosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) (1) • Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably (1) • Wood / biofuels are renewable but depletable if forests are felled faster than they regrow (1)	3
04.2	<i>1 mark for each point.</i> • Other factors (technology, location, cost) restrict how much can be harnessed (1) • Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures (1) • New technologies or higher prices may make these usable in future (1)	3
04.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid (1) • HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock (1) • Solar: high light intensity, low cloud cover (1) • Wave: coasts with strong reliable prevailing winds and a long fetch (1) • Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow (1) • Geothermal: hot rocks near the surface / recent volcanic activity (1) • Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
05.1	<i>1 mark for each point.</i> • Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged (1) • Electrons flow through an external conductor / circuit to the relatively positive lower layer; the current powers appliances (1)	2
05.2	<i>1 mark for each point.</i> • Smooth cells reflect about 30% of the light; grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away (1) • Multi-junction cells have several layers of different materials, each absorbing different wavelengths (1) • So a greater proportion of the available light is absorbed and converted to electricity (1)	3
05.3	<i>1 mark for each point.</i> • Solar farm: polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m² matters less) (1) • Satellite: multi-junction - highest efficiency where mass / area is critical and cost is less important (1) • Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit (1)	3

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> - Intermittent: no output at night and low in winter (daily and seasonal cycles) - predictable (1) - Unpredictable: cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
06.1	<i>1 mark for each point.</i> - Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) (1) - Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) (1)	2
06.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $(200 - 25) / 200 \times 100$ (1) $= 87.5\%$ (1)	2
06.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) (1) - Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths (1) - Higher doses caused sterility / death (1) - After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) (1) - Breeding success recovered so the population rose (1)	4
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - DDT persists in soil / sediments / food chains for many years so exposure continued (1) - Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly (1) - Other pressures (habitat loss, persecution) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
07.1	<i>1 mark for each point.</i> 2020 has higher peak flow in the wet season (32 vs $22 \text{ m}^3 \text{ s}^{-1}$ in March) (1) 2020 has much lower dry-season flow (2 vs $9 \text{ m}^3 \text{ s}^{-1}$ in July / August) (1) - Flow is much more variable / seasonal in 2020; the river almost dries up (1)	3
07.2	<i>1 mark for each point.</i> - In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow (1) - With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk (1) - Little stored ice remains to melt in the dry season so flow falls very low (1)	3
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Water shortages for irrigation / drinking / HEP in the dry season (1) - Flooding in the wet season (1) - Sudden floods from ice-lake outbursts (glacial lake outburst floods) (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances (1) • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
08.1	<i>1 mark for each point.</i> • t-test (1) • The data are measured (continuous) values and two means are being compared (1) • The data are (assumed to be) normally distributed (1)	3
08.2	<i>1 mark.</i> • Use the Mann-Whitney U test instead (compares medians) (1)	1
08.3	<i>1 mark for each point.</i> • There is a 2% probability that the difference occurred by random chance (1) • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected (1)	2
08.4	<i>1 mark for each point.</i> • The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 (1) • About 68% of values lie within one SD of each mean so the two data sets are clearly separated (1)	2
08.5	<i>1 mark for each point.</i> • Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists (1) • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
09.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Volume of ash is much less than the original waste (less landfill) (1) • Heat used for district heating / electricity generation (energy recovery) (1) • No sorting / complicated management needed (1) • Destroys pathogens / flammable organic wastes converted to CO₂ and water (1) • Refuse-derived fuel can replace fossil fuels (eg cement kilns) (1)	3
09.2	<i>1 mark for each point.</i> • High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) (1) • Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH (1) • Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value (1) • Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost (1)	4

Question	Marking guidance	Marks
09.3	1 mark for each point. - Ash may contain concentrated heavy metals / toxic residues (1) - Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble (1)	2
09.4	Any 1 from the points below - 1 mark each, maximum 1. - Fuel (natural gas) needed to burn wet / non-flammable wastes (1) - Cost of treating effluent gases (SO₂, NO_x, HCl, smoke) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel - Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade - Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses - Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) - Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants - Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) - Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) - Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together - Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring - Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9

Question	Marking guidance	Marks
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction <i>and</i> a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	<i>1 mark.</i> • The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25